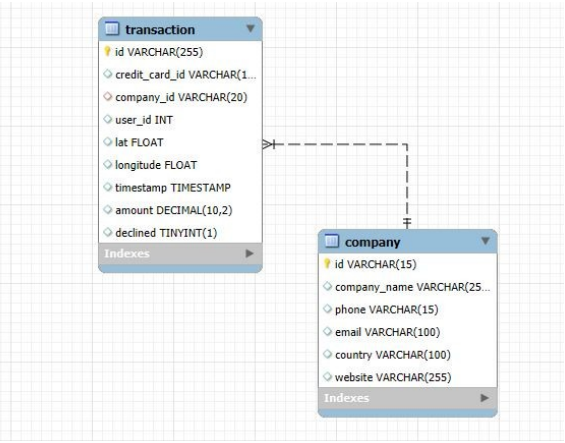
# SPRINT 2 FIAMMETTA ROFRANO

Nivel 1

Ejercicio 1

A partir de los documentos adjuntos (estructura_datos y datos_introducir), importa las dos tablas. Muestra las principales características del esquema creado y explica las diferentes tablas y variables que existen. Asegúrate de incluir un diagrama que ilustre la relación entre las distintas tablas y variables.

El esquema consta de dos tablas.
La tabla company proporciona los datos de distintas empresas (nombre, telefono, email, pais y website), el id representa la llave primaria.
La tabla transaction es descriptiva de distintas transacciones y se conecta a la tabla company mediante la clave foranea company_id.
En este caso la tabla company (dimention table) representa una tabla de dimension de transactions (fact table) ya que una empresa puede tener muchas transacciones en una relaccion de uno a muchos.




Ejercicio 2
Utilizando JOIN realizarás las siguientes consultas:

2.1 Listado de los países que están generando ventas.

```
SELECT DISTINCT country
FROM company AS c
JOIN transaction AS t
ON c.id = t.company_id
WHERE t.declined = 0
ORDER BY country;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| country |
|---|
| Australia |
| Belgium |
| Canada |
| China |
| France |
| Germany |
| Ireland |

Result 49

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 60 | 12:11:35 | SELECT DISTINCT country FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.declined = 0 ORDER BY country | 15 row(s) returned |

2.2 Desde cuántos países se generan las ventas.

```
SELECT COUNT(DISTINCT country) AS Paises_Generando_Ventas
FROM company AS c
JOIN transaction AS t
ON c.id = t.company_id
WHERE t.declined = 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Paises_Generando_Ventas |
|---|
| 15 |

Result 50

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 60 | 12:11:35 | SELECT DISTINCT country FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.declined = 0 ORDER BY country | 15 row(s) returned |
| 61 | 12:12:17 | SELECT DISTINCT country FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.declined = 0 ORDER BY country | 15 row(s) returned |
| 62 | 12:15:29 | SELECT COUNT(DISTINCT country) AS Paises_Generando_Ventas FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.d... | 1 row(s) returned |

2.3 Identifica a la compañía con la mayor media de ventas.

```
SELECT c.id,  c.company_name, ROUND(AVG(amount),2) AS media_de_ventas
FROM company AS c
JOIN transaction AS t
ON c.id = t.company_id
WHERE t.declined = 0
GROUP BY c.id, c.company_name
ORDER BY AVG(amount) desc  #  Mira el round en order by
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| id | company_name | media_de_ventas |
|---|---|---|
| b-2222 | Ac Fermentum Incorporated | 284.91 |

Result 51

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 61 | 12:12:17 | SELECT DISTINCT country FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.declined = 0 ORDER BY country | 15 row(s) returned |
| 62 | 12:15:29 | SELECT COUNT(DISTINCT country) AS Paises_Generando_Ventas FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.d... | 1 row(s) returned |
| 63 | 12:21:43 | SELECT c.id, c.company_name, ROUND(AVG(amount),2) AS media_de_ventas FROM company AS c JOIN transaction AS t ON c.id = t.company_id ... | 1 row(s) returned |

Ejercicio 3
Utilizando sólo subconsultas (sin utilizar JOIN):

3.1 Muestra todas las transacciones realizadas por empresas de Alemania.

```
SELECT *
FROM transaction AS t
WHERE t.declined = 0 AND company_id IN (SELECT id
                                        FROM company
                                        WHERE country = "Germany");
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined |
|---|---|---|---|---|---|---|---|---|
| 00138D3B-206D-4C03-94B7-63A2676EB9B4 | CcS-4899 | b-2222 | 318 | 41.3781 | 12.447 | 2020-03-25 10:43:43 | 426.36 | 0 |
| 0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9 | CcS-5070 | b-2222 | 489 | 41.3814 | 2.18176 | 2020-12-17 18:15:37 | 316.90 | 0 |
| 00201A11-2E62-44C4-941D-198FC8DB77F0 | CcU-3512 | b-2222 | 193 | 55.5704 | -3.65129 | 2021-01-22 23:44:27 | 453.04 | 0 |
| 00235618-0A5C-4D49-9DCB-B3A9405D8923 | CcS-8137 | b-2222 | 3556 | 59.8421 | 18.729 | 2020-09-09 15:43:19 | 263.14 | 0 |
| 005A5A7B-1F1A-4B6C-9B15-1625A78C9C38 | CcS-8998 | b-2222 | 4417 | 41.1591 | -8.63905 | 2024-05-15 09:10:11 | 442.01 | 0 |

transaction 54

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 66 | 12:28:54 | SELECT * FROM transaction AS t WHERE t.declined = 0 AND company_id IN (SELECT id FROM company WHERE country = "Germany") | 13269 row(s) returned |
| 67 | 12:29:03 | SELECT c.id, c.company_name, ROUND(AVG(amount),2) AS media_de_ventas FROM company AS c JOIN transaction AS t ON c.id = t.company_id ... | 1 row(s) returned |
| 68 | 12:29:11 | SELECT * FROM transaction AS t WHERE t.declined = 0 AND company_id IN (SELECT id FROM company WHERE country = "Germany") | 13269 row(s) returned |


| # | Time | Action | Message |
|---|---|---|---|
| 66 | 12:28:54 | SELECT * FROM transaction AS t WHERE t.declined = 0 AND company_id IN (SELECT id FROM company WHERE country = "Germany") | 13269 row(s) returned |
| 67 | 12:29:03 | SELECT c.id, c.company_name, ROUND(AVG(amount),2) AS media_de_ventas FROM company AS c JOIN transaction AS t ON c.id = t.company_id ... | 1 row(s) returned |
| 68 | 12:29:11 | SELECT * FROM transaction AS t WHERE t.declined = 0 AND company_id IN (SELECT id FROM company WHERE country = "Germany") | 13269 row(s) returned |

3.2 Lista las empresas que han realizado transacciones por un amount superior a la media de todas las transacciones

```
SELECT company_name
FROM company c
WHERE (SELECT AVG(amount)
        FROM transaction t
        WHERE t.declined = 0 AND t.company_id = c.id) > (SELECT ROUND(AVG(amount), 2)
                                                         FROM transaction AS t
                                                         WHERE t.declined = 0 );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name |
|---|
| Ac Fermentum Incorporated |
| Magna A Neque Industries |
| Fusce Corp. |
| Ante Iaculis Nec Foundation |
| Donec Ltd |

company 71

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 83 | 13:23:24 | SELECT * FROM transaction AS t JOIN company AS c ON c.id = t.company_id WHERE t.declined = 0 AND c.company_name != "Non Institute" AND ... | 12213 row(s) returned |
| 84 | 13:25:22 | SELECT company_name FROM company c WHERE (SELECT AVG(amount) FROM transaction t WHERE t.declined = 0 AND t.company_id = c.id) > (... | 48 row(s) returned |
| 85 | 13:25:51 | SELECT company_name FROM company c WHERE (SELECT AVG(amount) FROM transaction t WHERE t.declined = 0 AND t.company_id = c.id) > (... | 48 row(s) returned |

3.3Eliminarán del sistema las empresas que carecen de transacciones registradas, entrega el listado de estas empresas.

```
#Lista empresas
SELECT c.id, c.company_name
FROM company AS c
WHERE c.id NOT IN (SELECT DISTINCT t.company_id
                   FROM transaction AS t
                   WHERE t.declined = 0);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | company_name |
|---|---|
| NULL | NULL |

company 56

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 68 | 12:29:11 | SELECT * FROM transaction AS t WHERE t.declined = 0 AND company_id IN (SELECT id FROM company WHERE country = "Germany") | 13269 row(s) returned |
| 69 | 12:31:34 | SELECT company_name FROM company c WHERE (SELECT AVG(amount) FROM transaction t WHERE t.declined = 0 AND t.company_id = c.id) > (... | 48 row(s) returned |
| 70 | 12:57:46 | SELECT c.id, c.company_name FROM company AS c WHERE c.id NOT IN (SELECT DISTINCT t.company_id FROM transaction AS t WHERE t.decli... | 0 row(s) returned |

Nivel 2

Ejercicio 1

Identifica los cinco días que la mayor cantidad de ingresos en la empresa fue generada por las ventas.
Muestra la fecha de cada transacción junto con el total de ventas.

```
SELECT DATE(t.timestamp) AS fecha,
       SUM(t.amount) AS importe_total
FROM transaction AS t
WHERE t.declined = 0
GROUP BY fecha
ORDER BY importe_total DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| fecha | importe_total |
|---|---|
| 2022-12-13 | 14337.44 |
| 2019-11-18 | 13591.32 |
| 2023-02-20 | 13332.59 |
| 2017-12-20 | 13318.43 |
| 2019-03-18 | 12680.95 |

Result 57

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 69 | 12:31:34 | SELECT company_name FROM company c WHERE (SELECT AVG(amount) FROM transaction t WHERE t.declined = 0 AND t.company_id = c.id) > (... | 48 row(s) returned |
| 70 | 12:57:46 | SELECT c.id, c.company_name FROM company AS c WHERE c.id NOT IN (SELECT DISTINCT t.company_id FROM transaction AS t WHERE t.decli... | 0 row(s) returned |
| 71 | 13:04:21 | SELECT DATE(t.timestamp) AS fecha, SUM(t.amount) AS importe_total FROM transaction AS t WHERE t.declined = 0 GROUP BY fecha ORDER BY ... | 5 row(s) returned |

Ejercicio 2

¿Cuál es la media de ventas por país? Presenta los resultados ordenados de mayor a menor medio.

```
# Ejercicio 2 ¿Cuál es la media de ventas por país? Presenta los resultados ordenados de mayor a menor medio.

SELECT c.country, ROUND(AVG(t.amount), 2) AS media_ventas_pais
FROM company AS c
JOIN transaction AS t
ON c.id = t.company_id
WHERE t.declined = 0
GROUP BY c.country
ORDER BY AVG(t.amount) DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| country | media_ventas_pais |
|---|---|
| Australia | 265.54 |
| United States | 264.42 |
| Belgium | 260.97 |
| Germany | 260.83 |
| Ireland | 260.39 |

Result 60

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 72 | 13:09:14 | SELECT DATE(t.timestamp) AS fecha, SUM(t.amount) AS importe_total FROM transaction AS t WHERE t.declined = 0 GROUP BY fecha ORDER BY ... | 5 row(s) returned |
| 73 | 13:10:27 | SELECT c.country, ROUND(AVG(t.amount), 2) AS media_ventas_pais FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.... | 15 row(s) returned |
| 74 | 13:10:58 | SELECT c.country, ROUND(AVG(t.amount), 2) AS media_ventas_pais FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.... | 15 row(s) returned |

Ejercicio 3

En tu empresa, se plantea un nuevo proyecto para lanzar algunas campañas publicitarias para hacer competencia a la compañía "Non Institute". Para ello, te piden la lista de todas las transacciones realizadas por empresas que están ubicadas en el mismo país que esta compañía.

3.1 Muestra el listado aplicando JOIN y subconsultas.

```
SELECT *
FROM transaction AS t
JOIN company AS c
ON c.id = t.company_id
WHERE t.declined = 0 AND c.company_name != "Non Institute" AND c.country IN (SELECT country
                                                                              FROM company
                                                                              WHERE company_name = "Non Institute");
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined | id | company_name | phone | email | country | website |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 008629B4-C9A9-406C-A3D2-71FDA47BC546 | CcS-7063 | b-2246 | 2482 | 45.7666 | 4.83048 | 2015-07-30 12:12:42 | 486.44 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 00B72BA4-54A3-4B8E-B13F-2D57535AA17A | CcS-8475 | b-2246 | 3894 | 55.6212 | -3.7546 | 2017-10-26 22:08:26 | 414.06 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C | CcS-8700 | b-2246 | 4119 | 55.856 | -3.15783 | 2018-01-27 13:44:36 | 103.73 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 023FFCE8-E618-4938-BF56-C8DF80540ADD | CcS-7816 | b-2246 | 3235 | 46.3568 | 1.82755 | 2016-12-19 11:53:45 | 219.28 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 02683BEB-EF91-4564-957B-D6F1662AB7C5 | CcS-9471 | b-2246 | 4890 | 42.1332 | 12.396 | 2017-01-10 21:09:29 | 326.87 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |

Result 61

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 73 | 13:10:27 | SELECT c.country, ROUND(AVG(t.amount), 2) AS media_ventas_pais FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.... | 15 row(s) returned |
| 74 | 13:10:58 | SELECT c.country, ROUND(AVG(t.amount), 2) AS media_ventas_pais FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t.... | 15 row(s) returned |
| 75 | 13:13:02 | SELECT * FROM transaction AS t JOIN company AS c ON c.id = t.company_id WHERE t.declined = 0 AND c.company_name != "Non Institute" AND ... | 12213 row(s) returned |

3.2 Muestra el listado aplicando solo subconsultas.

```
# 3.2 Muestra el listado aplicando solo subconsultas.

SELECT *
FROM transaction AS t
WHERE company_id IN (
    SELECT id
    FROM company AS c
    WHERE t.declined = 0 AND c.company_name != "Non Institute" AND country = (SELECT country
                                                                              FROM company
                                                                              WHERE company_name = "Non Institute")
                                                                              );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined | id | company_name | phone | email | country | website |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 008629B4-C9A9-406C-A3D2-71FDA47BC546 | CcS-7063 | b-2246 | 2482 | 45.7666 | 4.83048 | 2015-07-30 12:12:42 | 486.44 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 00B72BA4-54A3-4B8E-B13F-2D57535AA17A | CcS-8475 | b-2246 | 3894 | 55.6212 | -3.7546 | 2017-10-26 22:08:26 | 414.06 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C | CcS-8700 | b-2246 | 4119 | 55.856 | -3.15783 | 2018-01-27 13:44:36 | 103.73 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 023FFCE8-E618-4938-BF56-C8DF80540ADD | CcS-7816 | b-2246 | 3235 | 46.3568 | 1.82755 | 2016-12-19 11:53:45 | 219.28 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |
| 02683BEB-EF91-4564-957B-D6F1662AB7C5 | CcS-9471 | b-2246 | 4890 | 42.1332 | 12.396 | 2017-01-10 21:09:29 | 326.87 | 0 | b-2246 | Sed Nunc Ltd | 02 62 64 73 48 | nibh@yahoo.org | United Kingdom | https://cnn.com/one |

Result 63

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 75 | 13:13:02 | SELECT * FROM transaction AS t JOIN company AS c ON c.id = t.company_id WHERE t.declined = 0 AND c.company_name != "Non Institute" AND ... | 12213 row(s) returned |
| 76 | 13:14:41 | SELECT * FROM transaction AS t WHERE company_id IN ( SELECT id FROM company AS c WHERE t.declined = 0 AND c.company_name !... | 12213 row(s) returned |
| 77 | 13:14:53 | SELECT * FROM transaction AS t JOIN company AS c ON c.id = t.company_id WHERE t.declined = 0 AND c.company_name != "Non Institute" AND ... | 12213 row(s) returned |

Nivel 3

Ejercicio 1

Presenta el nombre, teléfono, país, fecha y amount, de aquellas empresas que realizaron transacciones con un valor comprendido entre 350 y 400 euros y en alguna de estas fechas: 29 de abril de 2015, 20 de julio de 2018 y 13 de marzo de 2024. Ordena los resultados de mayor a menor cantidad.

```
SELECT c.id, c.company_name, c.phone, c.country, DATE(t.timestamp) AS fecha, t.amount
FROM company AS c
JOIN transaction AS t
ON c.id = t.company_id
WHERE  t.declined = 0 AND t.amount BETWEEN 350 AND 400 AND DATE(t.timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13")
ORDER BY t.amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| id | company_name | phone | country | fecha | amount |
|---|---|---|---|---|---|
| b-2566 | Aliquam PC | 01 45 73 52 16 | Germany | 2024-03-13 | 399.84 |
| b-2294 | Auctor Mauris Vel LLP | 08 09 28 74 14 | United States | 2018-07-20 | 399.51 |
| b-2402 | At Pede Corp. | 06 14 48 33 15 | Italy | 2015-04-29 | 390.69 |
| b-2566 | Aliquam PC | 01 45 73 52 16 | Germany | 2024-03-13 | 388.29 |
| b-2574 | Orci Adipiscing Limited | 03 18 00 77 81 | United Kingdom | 2018-07-20 | 373.71 |

Result 66

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 78 | 13:16:57 | SELECT c.id, c.company_name, c.phone, c.country, DATE(t.timestamp) AS fecha, t.amount FROM company AS c JOIN transaction AS t ON c.id = t.co... | 8 row(s) returned |
| 79 | 13:17:07 | SELECT c.id, c.company_name, c.phone, c.country, DATE(t.timestamp) AS fecha, t.amount FROM company AS c JOIN transaction AS t ON c.id = t.co... | 8 row(s) returned |
| 80 | 13:17:24 | SELECT c.id, c.company_name, c.phone, c.country, DATE(t.timestamp) AS fecha, t.amount FROM company AS c JOIN transaction AS t ON c.id = t.co... | 8 row(s) returned |

Ejercicio 2

Necesitamos optimizar la asignación de los recursos y dependerá de la capacidad operativa que se requiera, por lo que te piden la información sobre la cantidad de transacciones que realizan las empresas, pero el departamento de recursos humanos es exigente y quiere un listado de las empresas en las que especifiques si tienen más de 400 transacciones o menos.

```
SELECT t.company_id, c.company_name, COUNT(*) AS transacciones_totales,
CASE
    WHEN COUNT(*) > 400 THEN "> 400 transacciones"
    ELSE "< 400 transacciones"
END AS rating
FROM transaction AS t
JOIN company AS c
ON c.id = t.company_id
WHERE t.declined = 0
GROUP BY t.company_id, c.company_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_id | company_name | transacciones_totales | rating |
|---|---|---|---|
| b-2458 | Eget Tincidunt Dui Institute | 1522 | > 400 transacciones |
| b-2370 | Non Justo Corp. | 1476 | > 400 transacciones |
| b-2390 | Neque Tellus Imperdiet Corp. | 422 | > 400 transacciones |
| b-2230 | Fusce Corp. | 445 | > 400 transacciones |
| b-2266 | Mus Aenean Eget Foundation | 1563 | > 400 transacciones |

Result 68

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 80 | 13:17:24 | SELECT c.id, c.company_name, c.phone, c.country, DATE(t.timestamp) AS fecha, t.amount FROM company AS c JOIN transaction AS t ON c.id = t.co... | 8 row(s) returned |
| 81 | 13:21:18 | SELECT t.company_id, c.company_name, COUNT(*) AS transacciones_totales, CASE WHEN COUNT(*) > 400 THEN "> 400 transacciones" EL... | 100 row(s) returned |
| 82 | 13:21:27 | SELECT t.company_id, c.company_name, COUNT(*) AS transacciones_totales, CASE WHEN COUNT(*) > 400 THEN "> 400 transacciones" EL... | 100 row(s) returned |